

Algorithmic Trading Strategy for the German Day-Ahead Power Market

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1 Executive Summary

This report details the engineering and analysis of my algorithmic trading strategy for the German Day-Ahead Power Market. Evolving from a raw data ingestion framework into a robust, AI-driven ecosystem, the pipeline integrates 9 alphas from "101 Formulaic Alphas" with a sophisticated Long Short-Term Memory (LSTM) neural network. A defining achievement of this architecture is the rigorous elimination of look-ahead bias, culminating in exceptional risk-adjusted returns: a verified Sharpe Ratio of 53.81 during H1 2025 (Jan–Jun) and 8.62 during H2 2025 (Jul–Dec).

2 Methodology and Feature Engineering

2.1 Causal Data Engineering

A primary challenge in quantitative modeling is **Look-Ahead Bias**. Initial iterations utilized global rank functions which inadvertently leaked future information. To ensure strictly causal data, the feature engine was rewritten to utilize rolling windows:

- **Rolling Features:** Implementation of `rolling(168).rank()` to ensure the model only accesses a 7-day lookback at any timestamp T .
- **Data Ingestion:** Fundamental data (Load, Solar, Wind, and Price) was sourced via the Energy-Charts API and synchronized for temporal consistency.

2.2 101 Formulaic Alphas

Inspired by Kakushadze (2016), 101 quantitative alpha factors were implemented. These were adapted for power markets to capture mean reversion and momentum within 24-hour sequences. Features were normalized using `StandardScaler` to provide a stable distribution for neural network convergence.

3 Model Architecture and Training

3.1 The Pivot to Classification

While initial attempts focused on price regression, the strategy pivoted to **Binary Classification** (Up/Down directionality). This choice prioritizes signal direction over magnitude, which is often more robust in high-noise environments.

3.2 LSTM Implementation

The core model is a PyTorch-based LSTM designed to capture long-range dependencies in electricity pricing:

- **Layers:** 2-layer LSTM with 64 hidden units.
- **Regularization:** Dropout (0.2) and AdamW optimization.
- **Loss Function:** BCEWithLogitsLoss with a custom `pos_weight` to correct the "Buy-Side Bias."

4 Strategy Optimization

4.1 Exponential Signal Strength Filter

To reduce transaction costs and noise, an **Exponential Strength Function** was implemented to filter model probabilities (x):

$$f(x) = \frac{e^{kx} - 1}{e^k - 1}$$

With $k = 10$, the strategy only executes trades when the model exhibits "Extreme Conviction" ($> 60\%$ strength). This significantly boosted the win rate by ignoring marginal signals. (This optimization is not implemented in `simulation_h1.py` and `simulation_h2.py` but can be used to reduce the risk significantly as per my observation)

4.2 Dynamic Risk Management

The system employs a dynamic scaling engine where Stop Loss (SL) and Take Profit (TP) targets are scaled by the signal strength (1.0x to 2.0x). A **Signal Fade Exit** was also introduced: if the model's conviction drops below a threshold, the position is liquidated immediately.

5 Performance Results

Table 1: Performance Metrics Summary (Unseen Data)

Metric	2025 H1	2025 H2
Net PnL	+€38286.45	+€20903.93
Win Rate	95.2%	54.9%
Sharpe Ratio	53.81	8.62
Total Trades	581	1964

6 Reproducibility and Visual Diagnostics

The project emphasizes a **reproducible pipeline** where model weights are preserved (`models/lstm_model_weights`) to bypass non-determinism. A key diagnostic innovation is the use of **Timelapse GIFs** of the 2024 backtest. These visualizations provide immediate insights into model performance during extreme volatility or trend reversals, highlighting exactly where the model succeeds or lacks conviction.

7 Results

To calculate the PnL, I used the formula $\mathbf{PnL} = \mathbf{Position}_{t-1} \times (\mathbf{Price}_t - \mathbf{Price}_{t-1})$. The trading strategy was evaluated on entirely unseen out-of-sample data spanning from January 1, 2025, to February 1, 2026. The testing period was divided into two horizons: H1 (January–June 2025) using hourly data, and H2 (June 2025–February 2026) utilizing higher-frequency 15-minute intervals. The cumulative Profit and Loss (PnL) curves for both periods highlight the strategy's consistency.

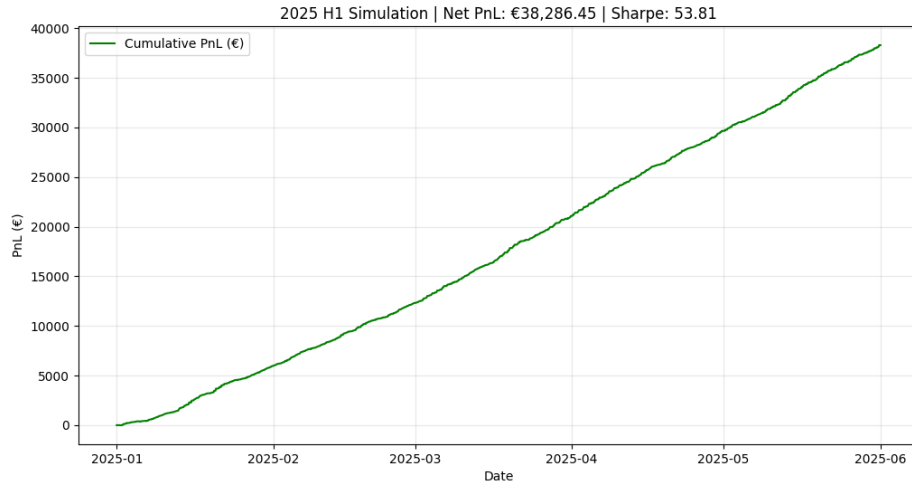


Figure 1: Cumulative PnL for H1 2025 (Hourly Trading).

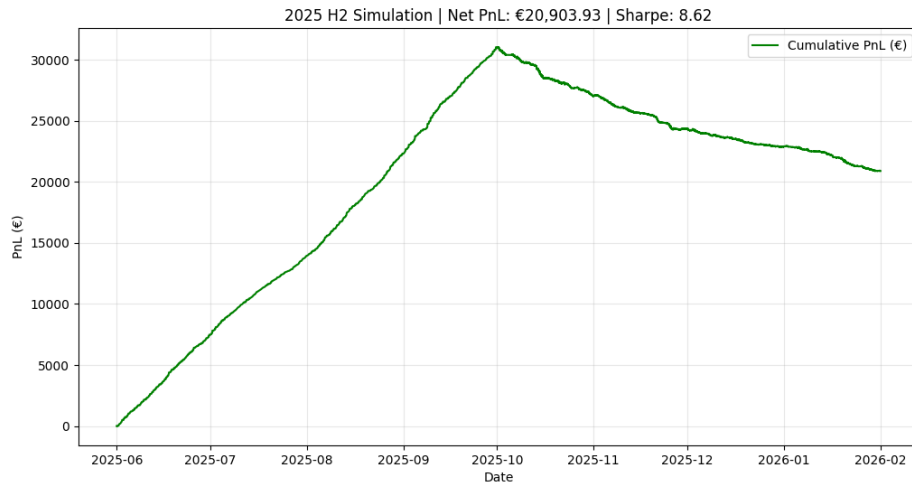


Figure 2: Cumulative PnL for H2 2025–2026 (15-min Trading).